Chapter 3 solutions

DM03-E01

For arbitrary x,

$$x \in A \cap (B \cup C) \iff x \in A \land (x \in B \lor x \in C)$$
$$\iff (x \in A \land x \in B) \lor (x \in A \land x \in C) \iff x \in (A \cap B) \cup (A \cap C).$$

The equivalences hold for every element, so extensionality gives equal sets.

DM03-E02

The individual sizes are 15,10,6. Intersections have sizes $\lfloor 30/6 \rfloor = 5$, 3,2, and the triple intersection contains only 30. Thus the union is

$$15 + 10 + 6 - 5 - 3 - 2 + 1 = 22.$$

Exactly-one regions give

$$(15 - 5 - 3 + 1) + (10 - 5 - 2 + 1) + (6 - 3 - 2 + 1) = 8 + 4 + 2 = 14.$$

DM03-E03

Sum seven has six outcomes, $(1, 6)$ through $(6, 1)$, among 36, so $P(A) = 1/6$. Given first die four, only second die three works, so $P(A|B) = 1/6$. Also $P(B) = 1/6$, hence

$$P(A \cap B) = 1/36 = P(A)P(B).$$

The events are independent.

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DM03-E04

Let D denote disease and $+$ a positive test. Then $P(D) = 0.02, P(+|D) = 0.95, P(+|D^c) = 0.10$, so

$$P(D|+) = \frac{0.95(0.02)}{0.95(0.02) + 0.10(0.98)} = \frac{0.019}{0.117} = \frac{19}{117} \approx 0.1624.$$

Under this assumed model, false positives constitute 9.8% of the population and true positives 1.9%. Expected counts for 10,000 people are:

Result	D	D^c	Total
Positive	190	980	1170
Negative	10	8820	8830
Total	200	9800	10000

Conditioning on positivity keeps the first row, giving $190/1170 = 19/117$. Sensitivity instead divides $190/200$ within the disease column. Prior probabilities determine the sizes of both columns, hence both contributions to the positive denominator. This table scales the given probabilities and introduces no new observations.

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DM03-E05

Normalization gives $6c = 1$, so $c = 1/6$. Then

$$EX = 0\frac{1}{6} + 1\frac{2}{6} + 2\frac{3}{6} = \frac{4}{3},$$
$$E[X^2] = 0\frac{1}{6} + 1\frac{2}{6} + 4\frac{3}{6} = \frac{7}{3},$$
$$\mathrm{Var}(X) = \frac{7}{3} - \left(\frac{4}{3}\right)^2 = \frac{5}{9}.$$

DM03-E06

No. With positive denominators,

$$P(A|B) > P(A) \iff P(A \cap B) > P(A)P(B) \iff P(B|A) > P(B).$$

The middle inequality is symmetric. The differences are

$$\frac{P(A \cap B) - P(A)P(B)}{P(B)}, \qquad \frac{P(A \cap B) - P(A)P(B)}{P(A)}.$$

Their positive denominators force the same sign. Less-than and equality comparisons also agree in direction.

DM03-E07

Since $A \cap B = \{b\}$,

$$\mathcal{P}(A) \cap \mathcal{P}(B) = \{\varnothing, \{b\}\}.$$

A subset of $A \cup B$ outside both powersets must contain c to fail inclusion in A, and a to fail inclusion in B. It may contain b or omit it. The difference is

$$\{\{a, c\}, \{a, b, c\}\}.$$

Generally, $A \subseteq B$ implies every $S \subseteq A$ lies in $\mathcal{P}(B)$, proving powerset inclusion. Conversely, if powersets are included, take arbitrary $a \in A$. Then $\{a\} \in \mathcal{P}(A) \subseteq \mathcal{P}(B)$, so $a \in B$. Thus $A \subseteq B$. If A is empty, the arbitrary-element argument holds vacuously.

DM03-E08

There are eight equally likely triples, and each equality event contains four, giving probability $1/2$. Any two equalities imply all three bits equal by transitivity. Every pair intersection is therefore $\{000, 111\}$, of probability $1/4 = (1/2)^2$. Pairwise independence holds.

The triple intersection is the same set, with probability $1/4$, not the marginal product $1/8$. Mutual independence fails. Two equalities already force the third, a higher-order restriction invisible in pairwise checks.

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DM03-E09

All size-three subsets number $\binom{12}{3} = 220$. Choosing defective and good objects gives

k	0	1	2	3
$\#$	56	112	48	4
$P(X=k)$	$14/55$	$28/55$	$12/55$	$1/55$

Counts sum to 220. Thus

$$EX = (28 + 24 + 3)/55 = 1, \qquad E[X^2] = (28 + 48 + 9)/55 = 17/11,$$

and variance is $17/11 - 1 = 6/11$.

For position indicators I_1, I_2, I_3, each has mean $1/3$ and variance $2/9$. Two positions are both defective with probability $(4/12)(3/11) = 1/11$, so covariance is $1/11 - 1/9 = -2/99$. Each of the three pairs occurs twice in the variance expansion:

$$\operatorname{Var}(X) = 3\frac{2}{9} + 2\binom{3}{2}\left(-\frac{2}{99}\right) = \frac{6}{11}.$$

Without-replacement dependence corrects variance, while expectation still adds directly.

DM03-E10

Bits	(I_1, I_2, I_3)	S
00	(0,0,0)	0
01	(0,1,1)	2
10	(1,0,1)	2
11	(1,1,0)	2

Thus $P(S=0) = 1/4, P(S=2) = 3/4$, with all other masses zero. Mean is $3/2$, second moment three, and variance $3 - 9/4 = 3/4$.

Binomial$(3, 1/2)$ has masses $1/8, 3/8, 3/8, 1/8$. Its mean $3/2$ and variance $3/4$ agree, but its PMF differs. Pairwise independence of the three indicators permits matching first and second moments. They are not mutually independent, so the binomial PMF does not follow. Two moments do not generally determine a distribution.

DM03-E11

Total probability gives

$$P(X=1)=\frac{2}{3}\frac{1}{4}+\frac{1}{3}\frac{3}{4}=\frac{5}{12}.$$

The marginal is Bernoulli$(5/12)$, with direct variance $35/144$. Both conditional variances are $3/16$, so

$$E[\mathrm{Var}(X|Y)]=3/16=27/144.$$

The conditional means are $m(0)=1/4, m(1)=3/4$, with total mean $5/12$. Their variance is

$$\frac{2}{3}\left(\frac{1}{4}-\frac{5}{12}\right)^2+\frac{1}{3}\left(\frac{3}{4}-\frac{5}{12}\right)^2=\frac{1}{18}=\frac{8}{144}.$$

The sum $27/144+8/144=35/144$ matches directly. Averaging within-group variance alone misses the between-group contribution $8/144$.

DM03-E12

Here $EX=0, EY=E[X^2]=2/3$. Since $XY=X^3$ has expectation zero, covariance is $0-0(2/3)=0$.

But $P(X=0,Y=0)=1/3$, whereas $P(X=0)P(Y=0)=1/9$, disproving independence. Zero covariance is only one second-moment condition; independence requires factorization of every joint mass.

Since $\mathrm{Var}\,X=2/3$ and Y is Bernoulli$(2/3)$ with variance $2/9$,

$$\mathrm{Var}(X+Y)=2/3+2/9+2(0)=8/9.$$

Directly, $X+Y$ is zero with probability $2/3$ and two with probability $1/3$. Its mean $2/3$ and second moment $4/3$ give the same $4/3-4/9=8/9$.

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DM03-E13

With $p = 1/3, q = 2/3$, mean is three and variance $q/p^2 = 6$. The exact tail is

$$P(T \geq 9) = P(T > 8) = (2/3)^8 = 256/6561.$$

Markov gives $P(T \geq 9) \leq ET/9 = 1/3$. Since $T \geq 9$ implies $|T - 3| \geq 6$, Chebyshev gives bound $6/36 = 1/6$. Both exceed the actual tail; Chebyshev is tighter here.

For nonnegative integers m, n, $P(T > m) = q^m > 0$, hence

$$P(T > m + n | T > m) = q^{m+n}/q^m = q^n = P(T > n).$$

This memoryless calculation uses $0 < p < 1$, ensuring positive conditioning tails. For $p = 1, m > 0$, the denominator would be zero.

DM03-E14

Each marginal is uniform on $\{0, 1\}$. Thus

$$P(X = 1|Y = 1) = \frac{1/8}{1/2} = 1/4,$$

which differs from the marginal $1/2$, so independence fails. Since $E[XY] = 1/8$, covariance is $1/8 - 1/4 = -1/8$.

The sum has masses $1/8, 6/8, 1/8$ at 0,1,2. Its mean is one, second moment $6/8 + 4/8 = 5/4$, and variance $1/4$. Covariance confirms

$$\mathrm{Var}(X + Y) = 1/4 + 1/4 + 2(-1/8) = 1/4.$$

The correction explicitly captures the joint dependence.

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DM03-E15

Use $1/[n(n+1)] = 1/n - 1/(n+1)$. The first N masses sum to $1 - 1/(N+1) \to 1$, and each is nonnegative. This is a valid countable probability distribution. The same telescoping gives, for positive integers m,

$$P(X \geq m) = \sum_{n=m}^{\infty} (1/n - 1/(n+1)) = 1/m.$$

However,

$$EX = \sum_{n=1}^{\infty} \frac{n}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n+1} = +\infty.$$

The second moment diverges as well, since each term is at least $1/2$:

$$E[X^2] = \sum_{n=1}^{\infty} \frac{n}{n+1} = +\infty.$$

For example, group terms by $2^j \leq n+1 < 2^{j+1}$; each group contributes a fixed positive lower bound, so the first-moment sum diverges. The nonnegative variable has a well-defined extended expectation but no finite mean. The finite-variance formula's premises fail, and infinity minus infinity is undefined.

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DM03-E16

The event $B = \{c\}$ is nonempty but has probability zero. It is a nonempty null event, distinct from the empty set.

Conditioning here requires $P(B) > 0$. Both numerator and denominator are zero, so $P(A|B)$ is undefined rather than zero.

Independence uses no division:

$$P(A \cap B) = 0 = P(A)P(B).$$

Thus A, B are independent. Every null event is independent of every event, since its intersections have zero mass by monotonicity. This does not conflict with positive-probability disjoint events being dependent; that claim assumes both probabilities positive.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 3.

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